SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ENG301

Examination - July Semester 2023

Microprocessor Programming

Tuesday, 14 November 2023 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (04)** questions and comprises **NINE (09)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Please detach Appendix and attach to Answer Booklet for submission.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (100 marks)

Question 1

(a) A 16-bit little endian computer has its 16-bit data stored as bytes in 8-bit memory as shown below.

Memory address	Data stored (hex)
0x1000	67
0x1001	45

A 16-bit big endian computer has a 16-bit value stored in its memory which is 127_{10} more than the above value. Appraise the resulting big endian 16-bit data in the below 8-bit memory location(s).

Memory address	Data stored (hex)
0x2000	
0x2001	

(3 marks)

(b) The MU0 processor has a 16-bit instruction set in which, the most significant 4 bits of the instruction define the opcode and the least significant 12 bits define the memory address of the operand data. The instruction set for the MU0 processor is provided below.

Instruction	Opcode	Effect
LDA S	0000	ACC := mem_{16}[S]
STO S	0001	mem_{16}[S] := ACC
ADD S	0010	ACC := ACC + mem_{16}[S]
SUB S	0011	ACC := ACC – mem_{16}[S]
JMP S	0100	PC := S
JGE S	0101	if ACC ≥ 0 PC := S
JNE S	0110	if ACC ≠ 0 PC := S
STP	0111	stop

Table Q1(b) given in Appendix A is a program code for the MU0 processor in which the 1st two columns contain the memory address and the MU0 instructions. Analyze the machine code instructions in Table Q1(b) and answer the below questions.

(i) Demonstrate your understanding of the machine code by filling the 3rd column of Table Q1(b) in Appendix A with the assembly language instruction corresponding to the machine code.

(11 marks)

(ii) Interpret each line of the code and provide your interpretation in the 4th column of Table Q1(b) in Appendix A.

(6 marks)

(iii) Appraise the final results of the program for the respective memory address(es).

(5 marks)

Question 2

The below assembly language code is run on an ARM processor with a 5-stage pipeline architecture which supports forwarding.

```
AND R1, R2, R3
ADD R2, R1, R2
LDR R1, [R2, #10]
SUB R2, R5, R1
STR R2, [R5, #20]
```

(a) Examine the above code and list all the data dependencies between the instructions in the above code.

(9 marks)

(b) Given that IF, ID, EX, MA, and WB represent fetch, decode, execute, memory access and write back pipeline stages, respectively. Fill in Table Q2(b) in Appendix B to show the pipeline stages while executing the above given code. Use "STALL" for bubble(s) that may arise, to ensure that the instructions execute without pipeline hazards.

(10 marks)

(c) Each of the 5 pipeline stages, i.e., IF, ID, EX, MA, and WB have a latency of 500 ps (pico seconds) and requires an additional 20 ps for the registers between the pipeline stages. Analyze and compute the cycle time, latency and throughput of the processor.

(6 marks)

Question 3

(a) The following program code is executed in an ARM processor:

```
MOVS    R7, #0XFFFFFFFE
MOVS    R5, #0
MOVS    R6, #-1
CMP     R5, R6
ADDEQS  R5, R7, R6
SUBNES  R5, R7, R6
```

Execute the above code and update Table Q3(a) given in Appendix C to show the values of the registers and flags after execution of each instruction.

(12 marks)

(b) Design an ARM assembly language program for the following pseudo code. You may use registers R0, R1 and R2 in ARM register file. Assume that the value "a" is available in R0 and the value 'b" is available in "R1". For each line of your assembly code, place a ";" at the end of each line and describe the code.

```
if (a >= 0 && b >= 0)
        return (a + b);
else if (a >= 0 && b < 0)
        return (a - b);
else if (a < 0 && b >= 0)
        return (b - a);
else
        return (a - b);
```

(13 marks)

Question 4

(a) Sketch the schematic block diagram of a simple memory interface for an embedded system using an ARM Cortex processor showing the control logic, four 8-bit ROM and four 8-bit RAM.

(7 marks)

(b) Analyze and explain how the ARM memory interface will provide address decoding to specific locations of RAM and ROM to access data as byte, half word and word.

(8 marks)

(c) Sketch the memory map for the ARM memory interface if the ROM used has a capacity of 512 MB and the RAM has a capacity of 4 MB.

(10 marks)

APPENDIX A

Student's PI no: _____

Instruction: Write your PI number and attach this page to answer sheets for submission.

Question 1(b):

Memory Address	MU0 instruction machine code / data (hex)	Assembly Mnemonic	Interpretation of the machine code
0x00	0010		
0x01	2012		
0x02	1010		
0x03	0011		
0x04	2012		
0x05	1011		
0x06	0013		
0x07	3014		
0x08	1013		
0x09	6000		
0x0A	7000		
0x0B	….		
0x0C	….		
0x0D	….		
0x0E	….		
0x0F	….		
0x10	0002		
0x11	0003		
0x12	0002		
0x13	0005		
0x14	0001		

Table Q1(b)

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APPENDIX B

Student's PI no: _____

Instruction: Write your PI number and attach this page to answer sheets for submission.

Question 2(b):

Clock Cycle → Instruction ↓	CC1	CC2	CC3	CC4	CC5	CC6	CC7	CC8	CC9	CC10	CC11	CC12
AND R1, R2, R3												
ADD R2, R1, R2												
LDR R1, [R2, #10]												
SUB R2, R5, R1												
STR R2, [R5, #20]												

Table Q2(b)

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APPENDIX C

Student's PI no: _____

Instruction: Write your PI number and attach this page to answer sheets for submission.

Question 3(a):

Assembly language Instruction	Register R5	Register R6	Register R7	N flag	Z flag	C flag	V flag	Q flag
MOVS R7, #0XFFFFFFFE								
MOVS R5, #0								
MOVS R6, #-1								
CMP R5, R6								
ADDEQS R5, R7, R6								
SUBNES R5, R7, R6								

Table Q3(a)

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